

A 2-D Vernier Time-to-Digital Converter

15 ps resolution, 5-bit thermometer output, TSMC 180 nm BCD

EE4615 Digital IC Design II | Group 22

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The brief: quantize a time interval to 5 bits

- **Task:** convert the delay between a START and a STOP edge into a 5-bit thermometer code $Q_1..Q_{31}$
- **Hard requirements:** resolution better than 20 ps, full 32-code range, and validated robustness in all five process corners
- **Rules of the game:** TSMC 180 nm BCD at 1.8 V, every gate hand-built (no standard cells), schematic plus Spectre only
- **Optimize the trade-off:** linearity (DNL, INL), energy per conversion, area as total transistor width ΣW , and FoM = energy per step
- **Our choice:** a two-dimensional Vernier TDC (Vercesi et al., JSSC 2010), the structure that decouples resolution from range

5 bit

thermometer output

< 20 ps

required resolution

5 / 5

corners must pass

0

standard cells allowed

Why Vernier, and why a 2-D grid

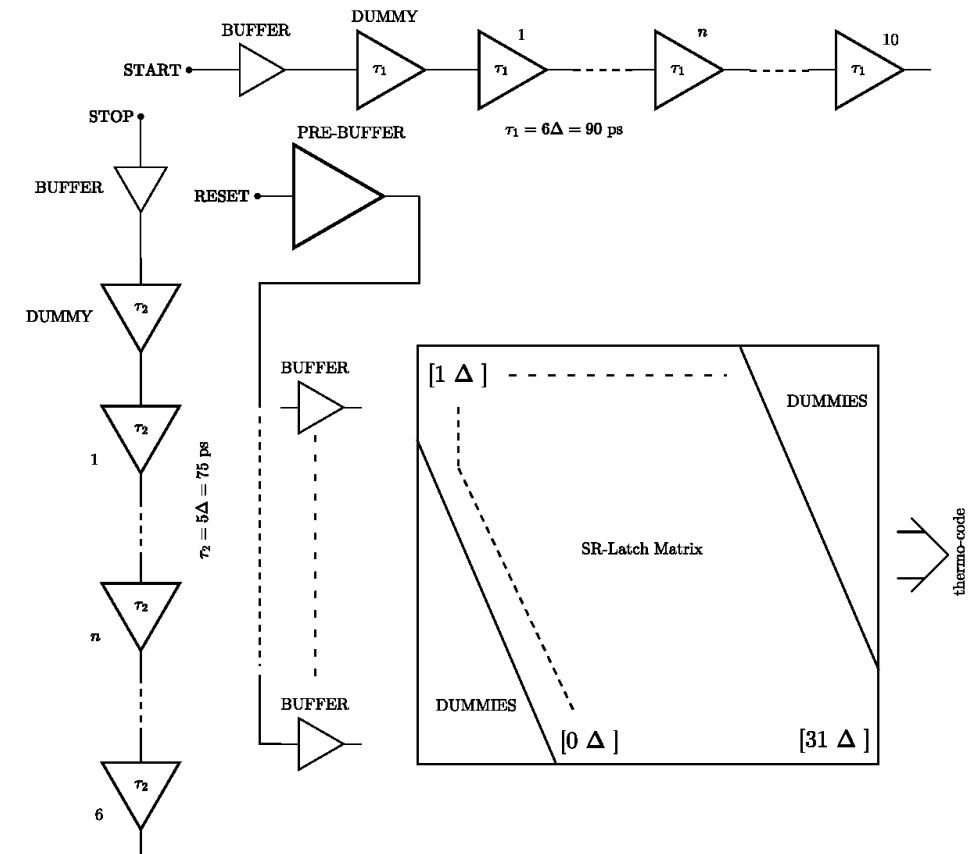
- **Vernier principle:** START runs through slow stages τ_1 , STOP chases through fast stages τ_2
 - the LSB is the difference $t_0 = \tau_1 - \tau_2$, far below a single gate delay
- **1-D catch:** range = $N \cdot t_0$, so 32 codes need 32 stages in each line; cells and latency grow linearly
- **2-D idea:** compare START tap i against STOP tap j ; every pair measures $\Delta t(i,j) = i \cdot \tau_1 - j \cdot \tau_2$
 - a grid of $N+N$ stages yields $\sim N^2/2$ distinct codes: range without long lines
- **Payoff for our 5-bit target:** 31 codes from only 10 + 6 delay stages, one SR-latch arbiter per code

For 32 codes	1-D Vernier	2-D (ours)
Delay stages	32 + 32	10 + 6
Arbiters	32	31 + 29 dummy
Full-scale latency	~ 3.3 ns	~ 0.9 ns
LSB	$\tau_1 - \tau_2$	$\tau_1 - \tau_2$

Same resolution, a quarter of the delay cells

Architecture: a 10×6 grid locked at $k = 6$

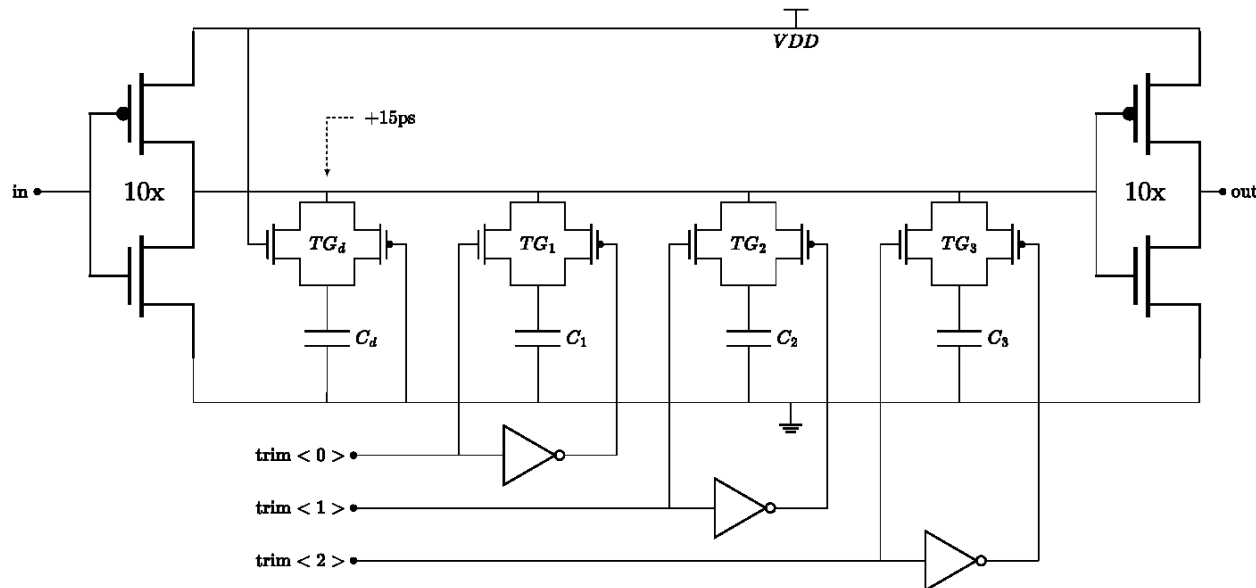
- **Operating point:** $\tau_1 = 6 \cdot t_0 = 90$ ps (10 stages), $\tau_2 = 5 \cdot t_0 = 75$ ps (6 stages), so $t_0 = 15$ ps with 5 ps of spec margin
- **Bijective routing:** each thermometer level m maps to exactly one grid cell, so no OR-tree and no skew from combining logic
 - $y_m = ((m-1) \bmod 6) + 1$, $x_m = (m + 5 \cdot y_m) / 6$
- **Matched loading:** 31 active SR latches plus 29 dummies pad the array to 10×10 , every tap drives identical fan-out
- **I/O:** buffer trees distribute START, STOP and the 40 ps RESET; the code is read straight off the latch outputs



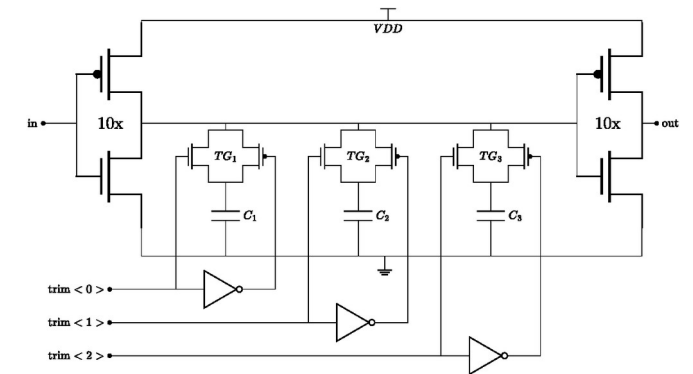
Two delay lines, SR-latch matrix, dummy padding

Delay stage: FO = 1 inverter pair with MOSCAP trim

- **Breaking the FO = 4 habit:** classical sizing pinned the stage floor at 97 ps; an FO = 1 pair opened a 74 to 147 ps tuning window
- **Twin cells:** both lines use identical active devices; the 6:5 delay ratio comes from loading, so τ_1 and τ_2 track across PVT
- **Calibration built in:** a fixed MOSCAP sets the nominal delay, three transmission-gate switched banks re-center t_0 per corner (authority ≈ 0 to 48 ps)



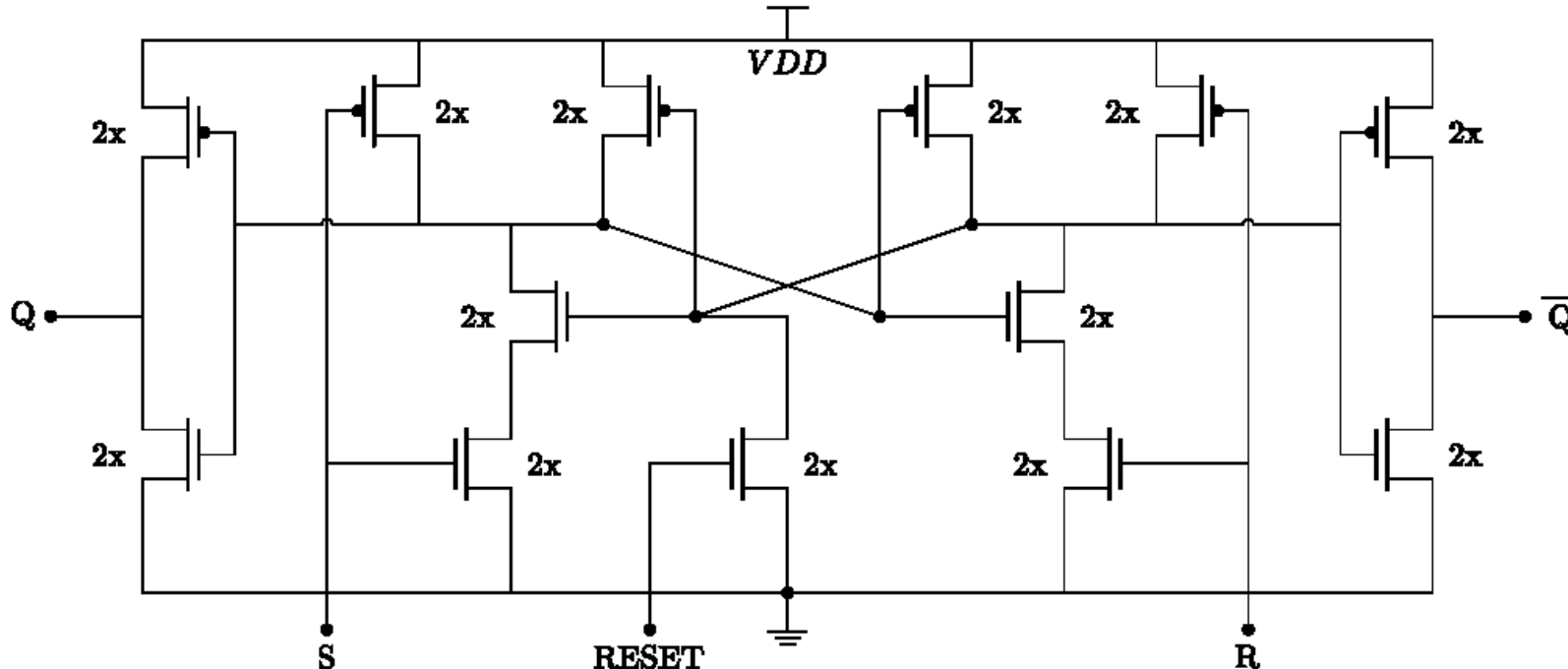
delay_tau1 (START, 90 ps): trim<0:2> banks behind transmission gates



delay_tau2 (STOP, 75 ps)

Arbiter: cross-coupled NAND SR latch, 13 transistors

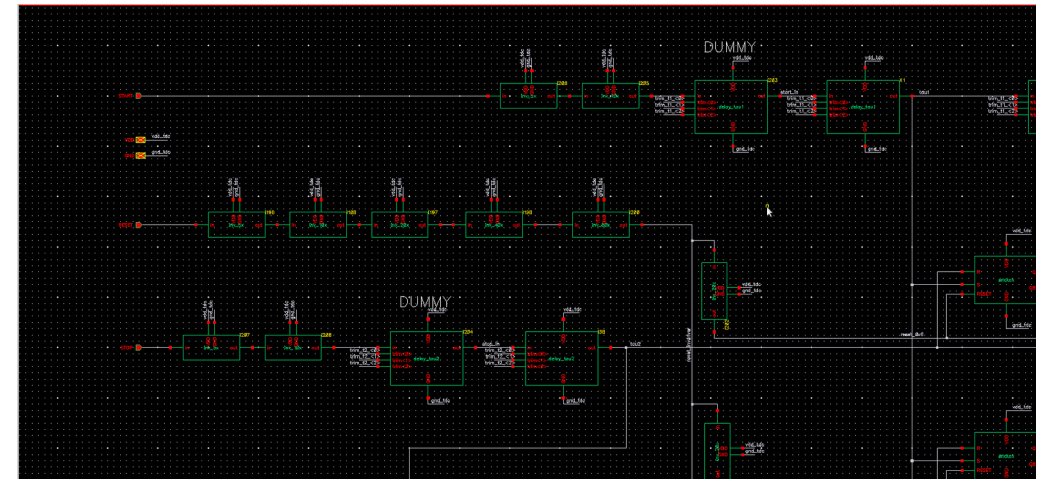
- **Topology:** two 2-input NANDs (all devices 440 nm / 180 nm), an async-reset NMOS, and buffered Q / $\bar{Q}$ outputs
- **Race logic:** both NAND outputs idle high; the first rising input wins, so the latch directly records which edge arrived first
- **Why it works:** equal N and P widths balance the S and R paths, keeping the arbitration offset at the femtosecond level
- **Why not a D flip-flop:** 60 instances make cell economy count; the SR latch is smaller, symmetric, and resets in one transistor



vernier2d/srlatch: S and R race, RESET clears the grid in 40 ps

The pivot: size against the real load, not the ideal one

- **The trap:** a grid tap drives a full row or column of latches plus the next delay stage, about a 20× equivalent load
- **We measured it:** a 1-D row built from the same cells with raw tap loading predicted $t_0 = 16.6$ ps, but measured 28.6 ps
 - sizing against an idealized load gives a fictitious LSB
- **The fix:** characterize every stage under the replica 20× load, and equalize all taps with dummy latches
- **Consequence:** the original $k = 4$ point (60 / 45 ps) was unreachable; the design re-locked at $k = 6$ (90 / 75 ps)
- **Lesson:** in a Vernier grid, loading is part of the architecture, not a layout afterthought

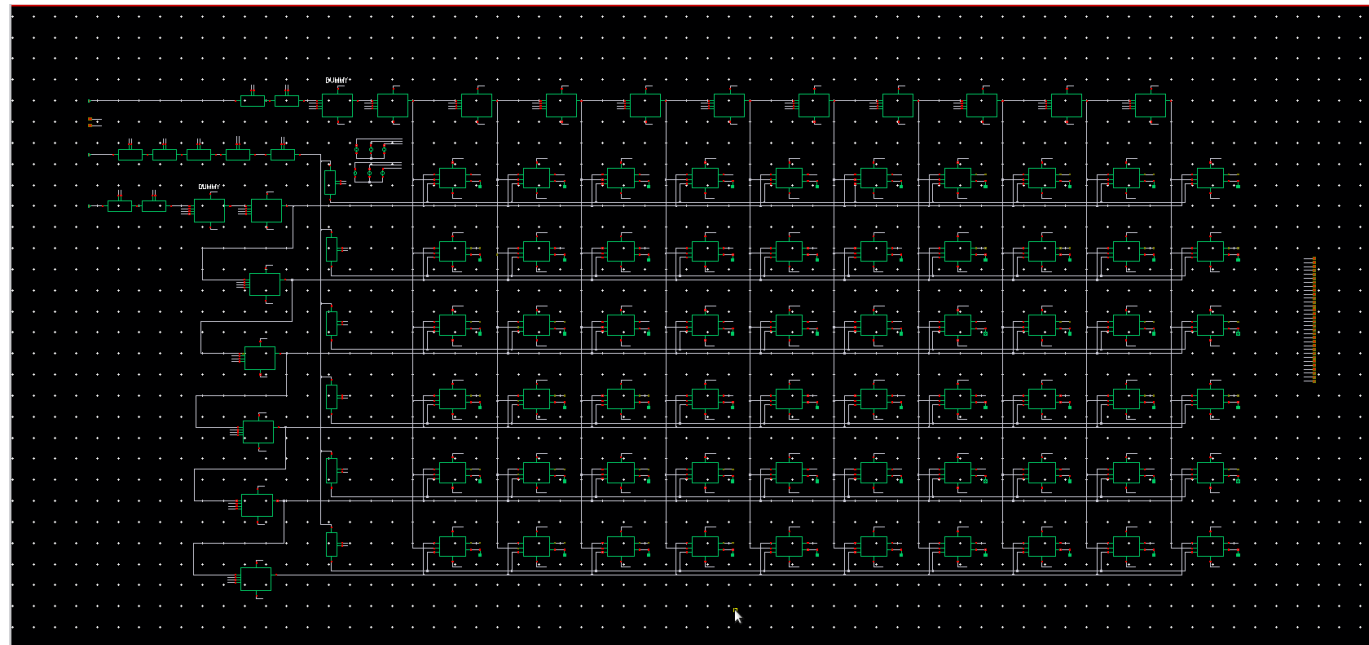


Grid detail: DUMMY latches pad every tap to fan-out 10

Built in Cadence, verified with the course testbench

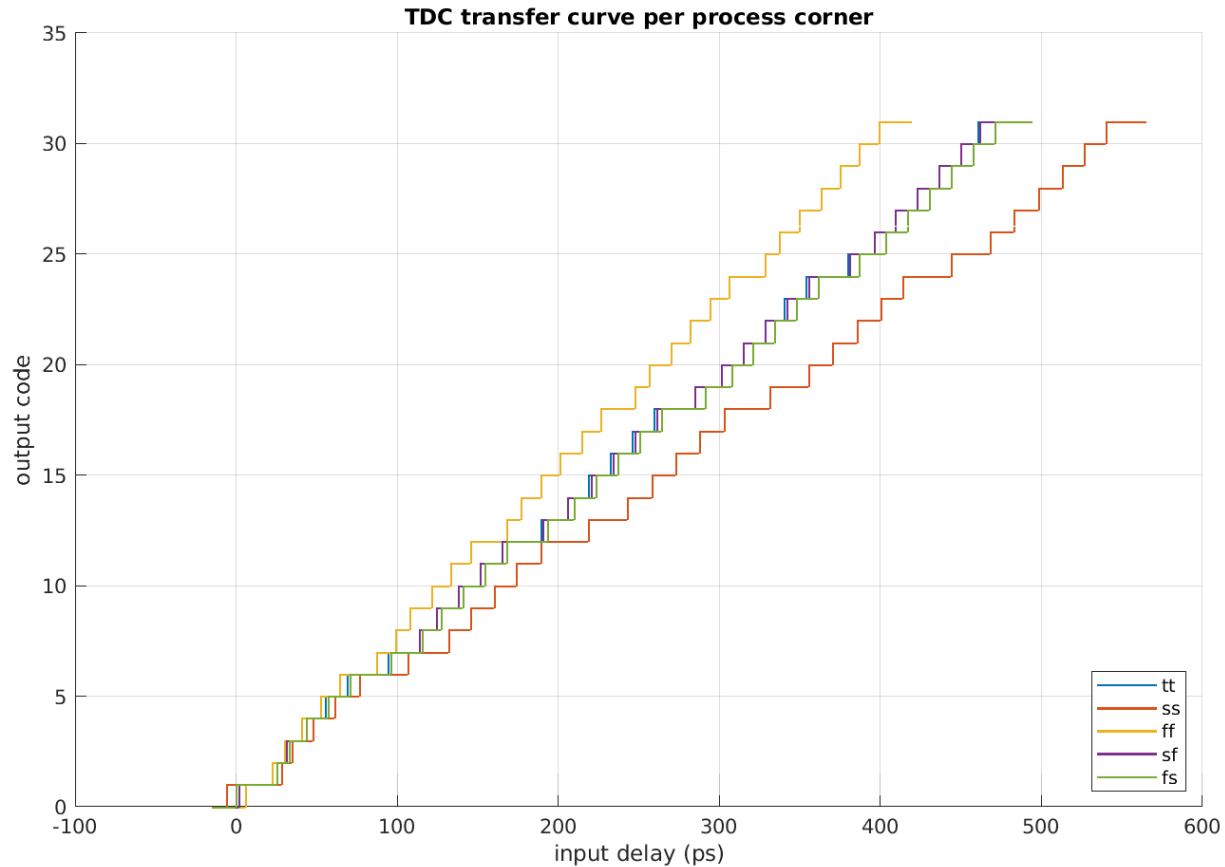
Daniel

- **The core:** 11 + 7 delay stages, 60 SR latches, 1120 devices, $\Sigma W = 1306 \mu\text{m}$, all cells hand-built in vernier2d
- **Honest energy numbers:** the DUT sits on its own supply behind a series ammeter; stimulus buffers run from a separate rail
- **Automated sweeps:** OCEAN steps Δt through all 32 codes per corner and extracts LSB, DNL, INL, energy and pass / fail



vernier2d/tdc_core: τ_1 line (top), τ_2 line (left), 10 × 10 padded latch matrix

Transfer: a clean 5-bit staircase in every corner



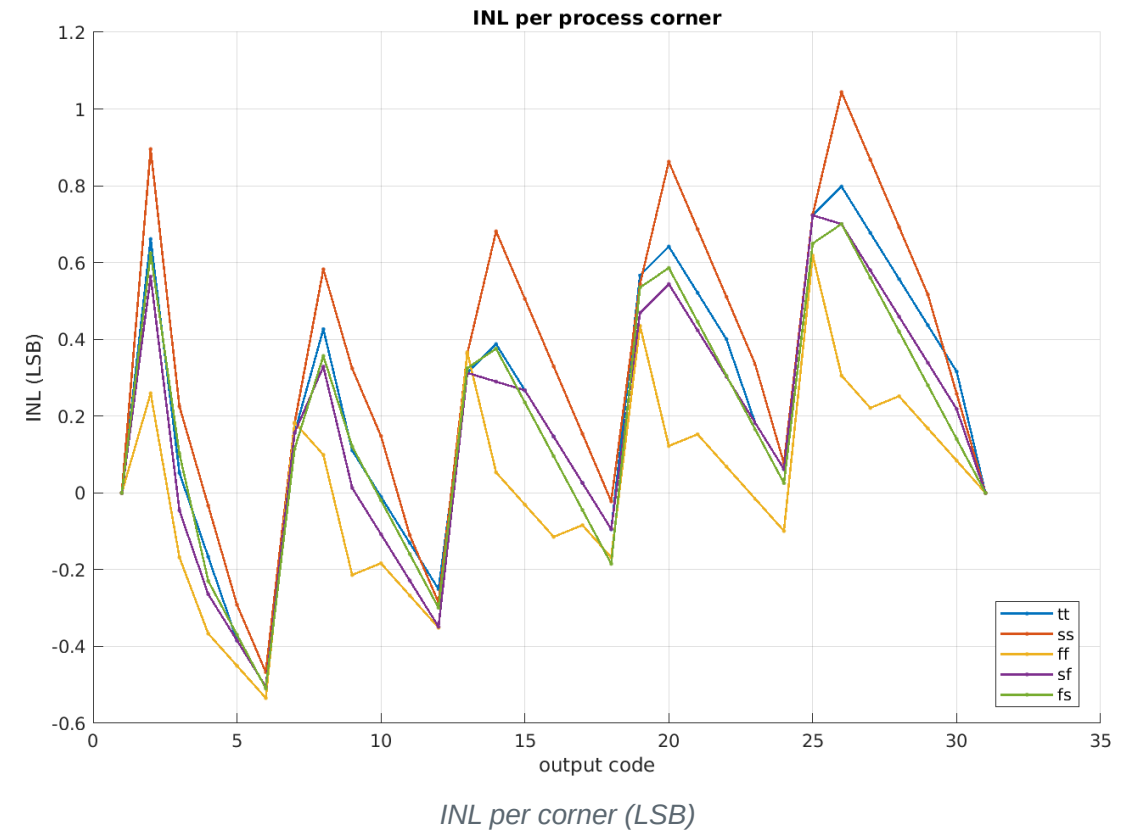
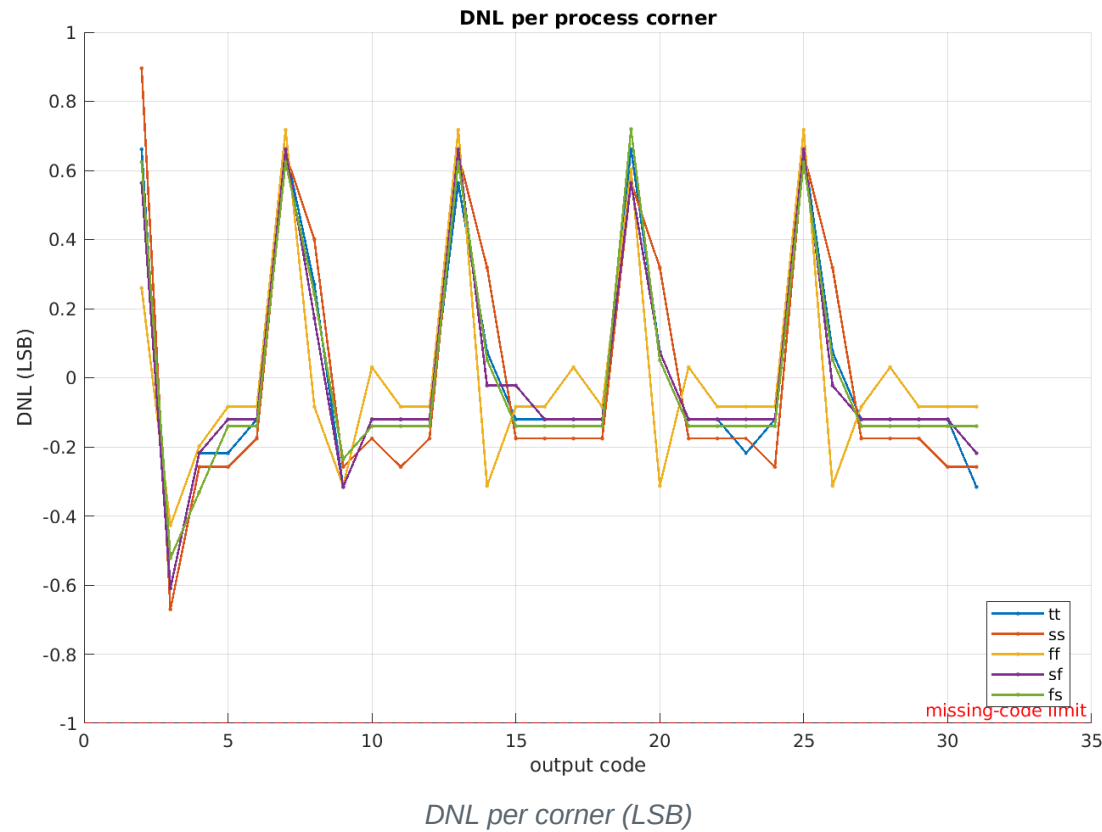
Measured TDC transfer, all corners, 27 °C, 1.8 V

Corner	LSB (ps)	Spec < 20 ps
TT	15.35	PASS
SS	18.20	PASS
FF	13.10	PASS
SF	15.35	PASS
FS	15.70	PASS

- One fixed trim code, no per-corner tweaking
- All 31 steps present in all five corners
- Offset stays a static, calibratable shift

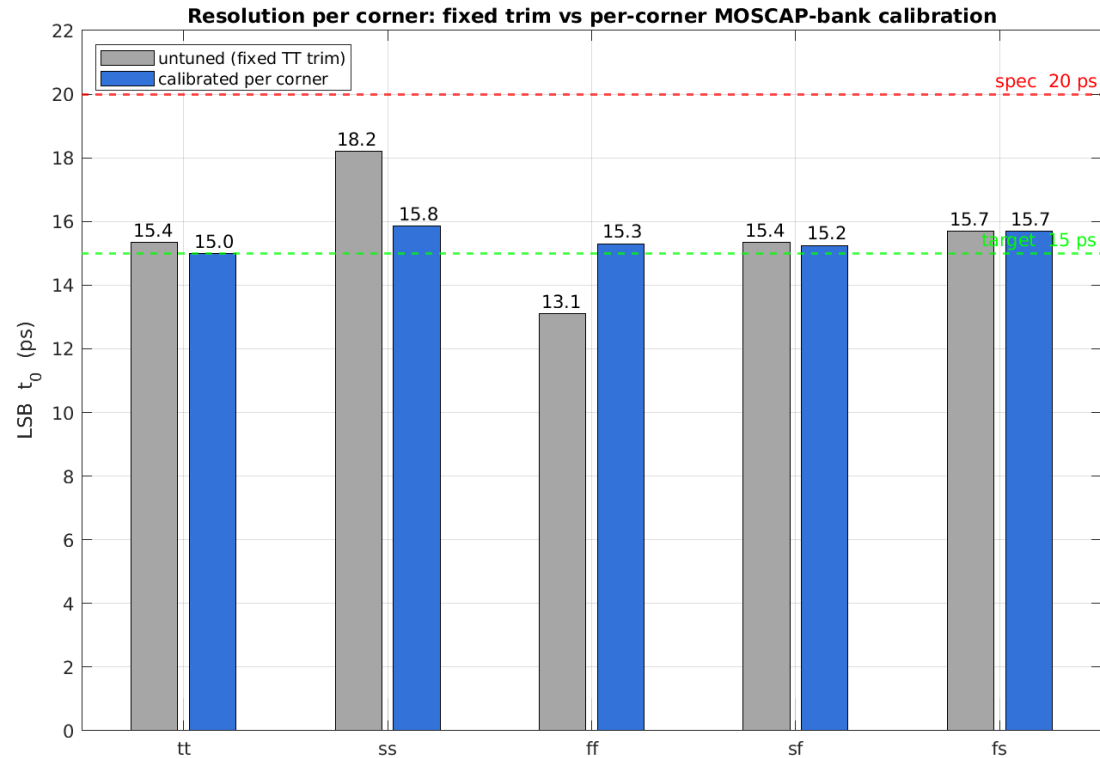
Linearity: no missing codes anywhere

Daniel

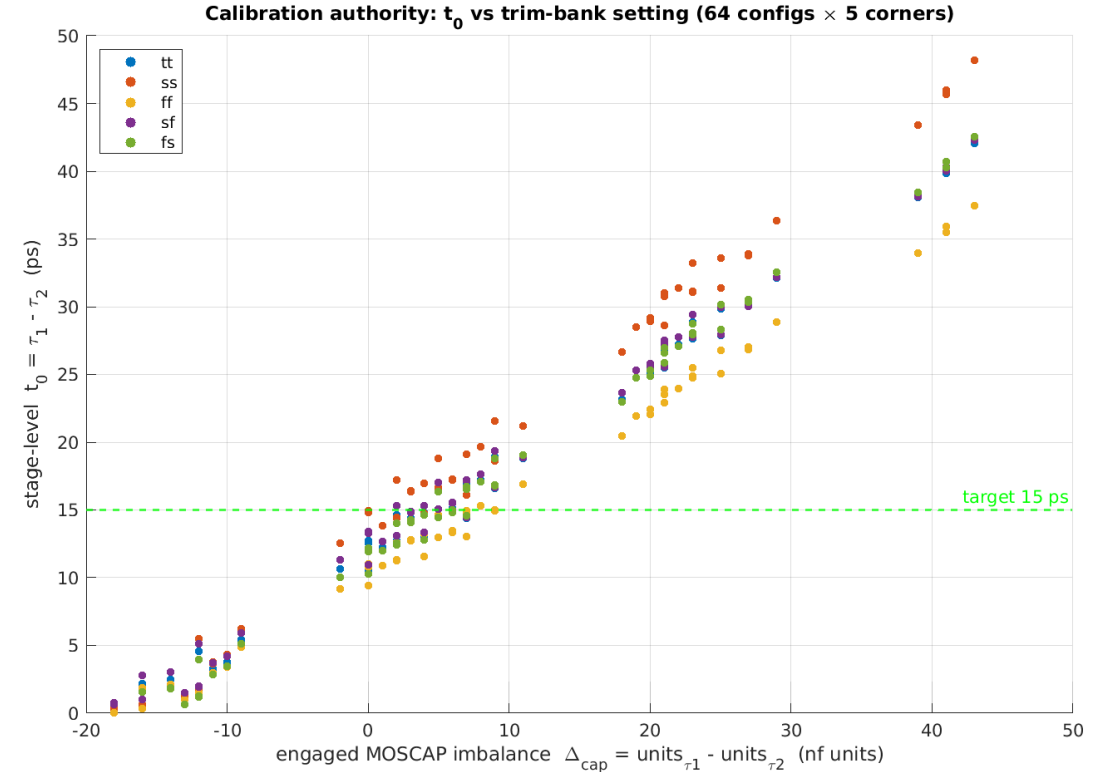


- Worst DNL 0.90 LSB (SS), far from the -1 LSB missing-code limit; worst INL 1.04 LSB, all other corners below 0.8
- The every-6-codes DNL signature is the row-to-column residual of the grid, the expected 2-D Vernier fingerprint

Per-corner MOSCAP calibration: 6× tighter LSB



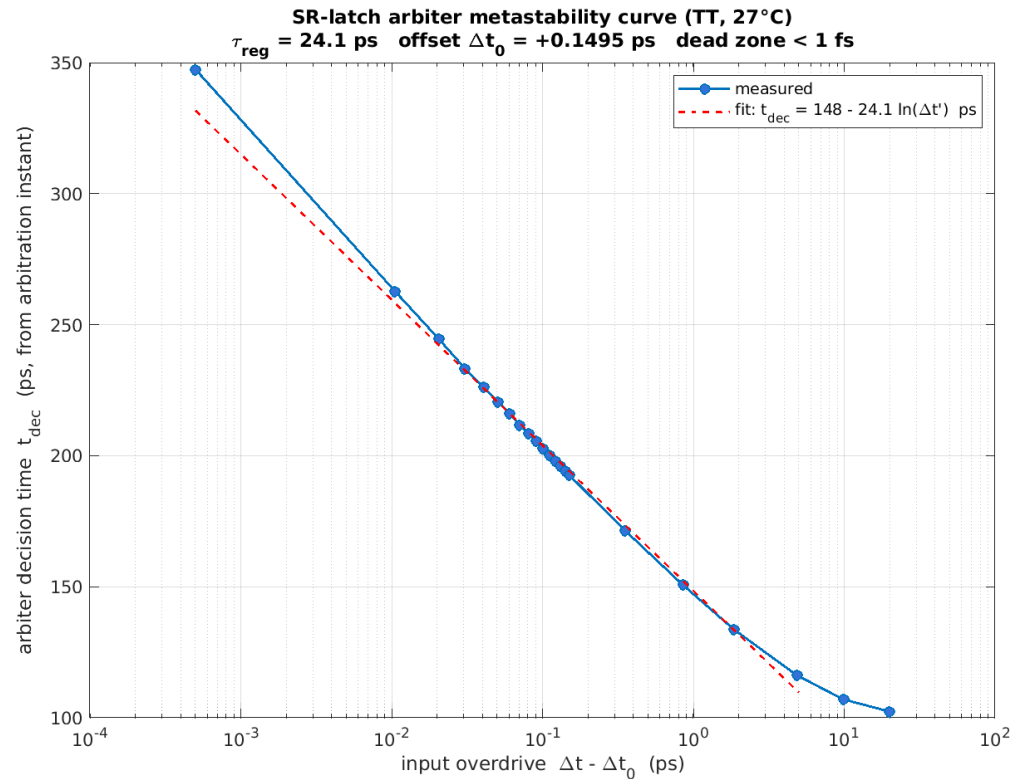
LSB per corner, untuned vs calibrated



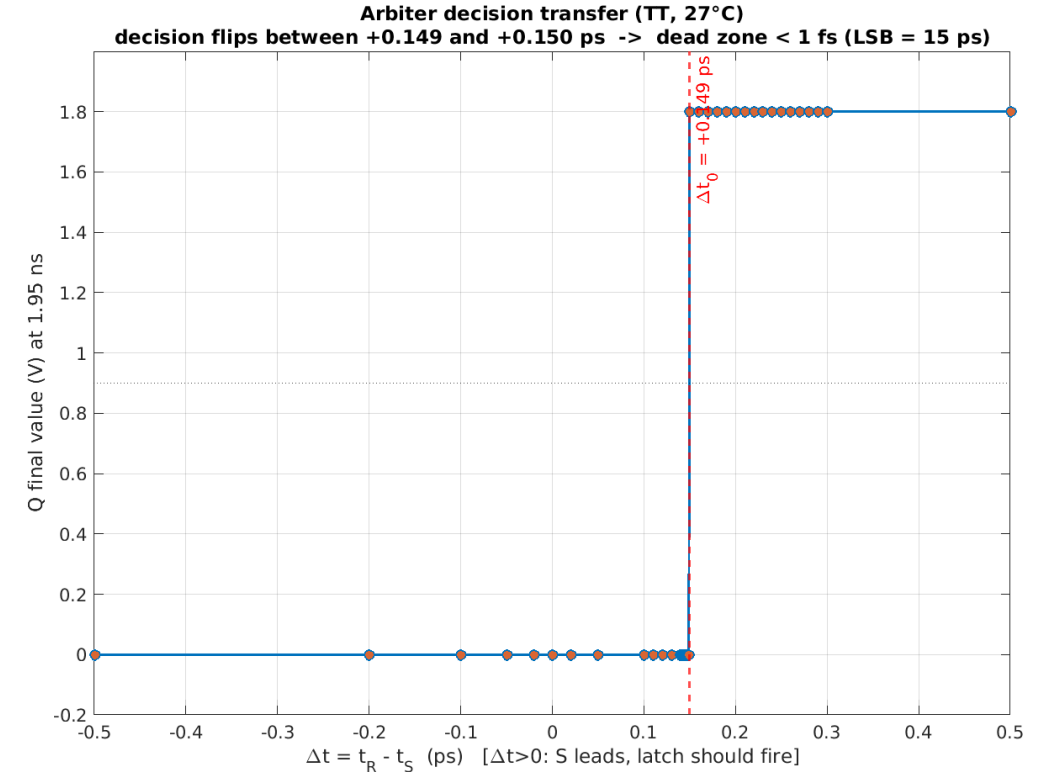
Trim-bank authority: t_0 vs configuration

- Three trim banks per stage act as static configuration bits, set once per corner
- LSB corner spread shrinks from 5.10 ps to 0.85 ps; every corner re-centers to 15.0 – 15.9 ps, worst INL drops to 0.74 LSB

How small a Δt can the arbiter resolve?



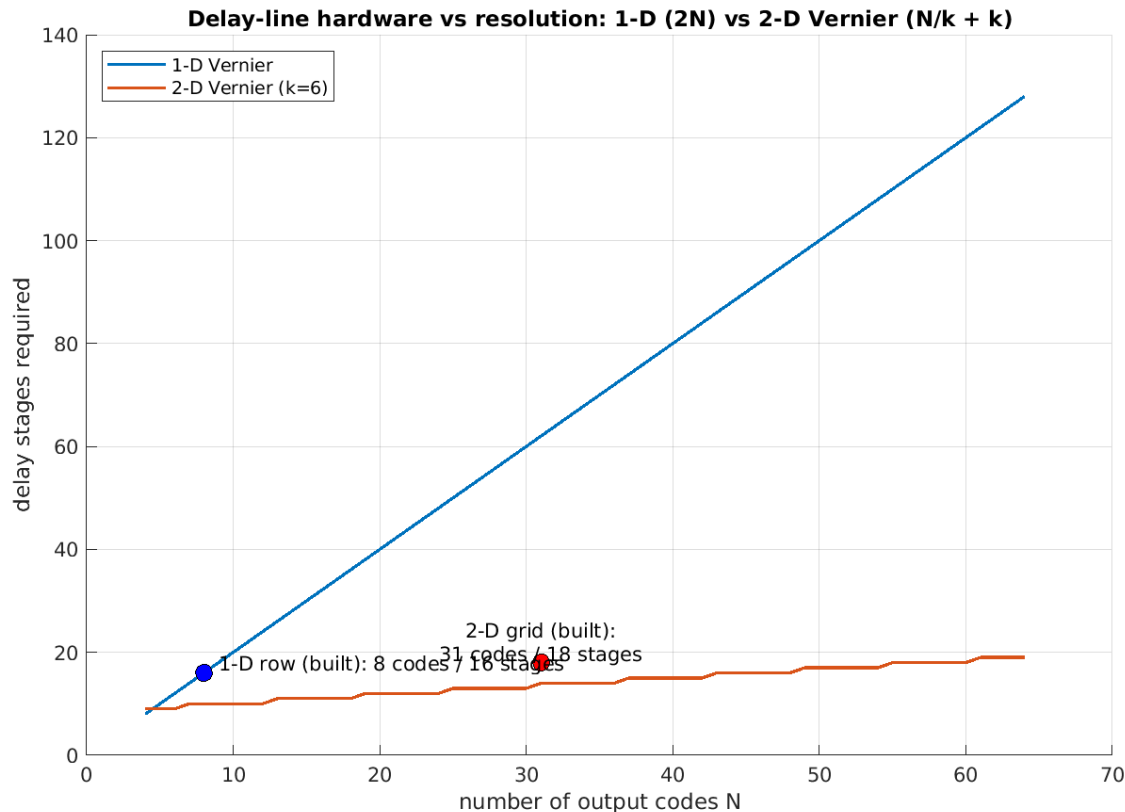
Decision time vs overdrive: $\tau = 24.1 \text{ ps}$ regeneration



Arbiter transfer: dead zone below 1 fs

- Arbitration offset +0.15 ps (1 % of one LSB); decision ambiguity below 1 fs, four orders under the 15 ps LSB
- Worst observed decision time 348 ps at 0.5 fs overdrive: metastability does not limit this design, the LSB does

Measured: 2-D beats 1-D built from the same cells



Cost scaling with code count, 1-D vs 2-D

Same cells, measured	1-D row	2-D core
LSB (ps)	28.57	15.35
DNL max (LSB)	0.89	0.66
ΣW per code (μm)	81.4	42.1
Energy per code (pJ)	0.327	0.376
Latency, 31 codes	3.3 ns	0.9 ns

- Raw tap loading costs the 1-D row almost 2× in resolution
- Scaling to 31 codes: 1-D needs 3.4× the stages and 1.9× the area

Scorecard: every requirement met, with margin

Joris

Requirement	Target	Achieved	Verdict
Output bits	5	5 (thermometer $Q_{1..Q_{31}}$)	PASS
Resolution (LSB)	< 20 ps	15.0 – 15.9 ps calibrated	PASS
Corner robustness	TT, SS, FF, SF, FS	5 / 5 corners pass	PASS
DNL	> -1 LSB (no missing codes)	worst 0.89 LSB	PASS
INL	report	worst 0.74 LSB (calibrated)	PASS
Energy per conversion	minimize	10.7 – 13.0 pJ	✓
FoM (energy per step)	minimize	0.48 – 0.53 pJ / step	✓
Area ΣW	minimize	1305.6 μm (1120 devices)	✓
ENOB	report	4.40 – 4.62 bit	✓

All numbers from Spectre corner sweeps at 27 °C, 1.8 V, course TB_TDC harness

Conclusions and outlook

What we showed

- A 15 ps, 5-bit 2-D Vernier TDC in 180 nm that meets every course spec in all five corners
- Two ideas carried the design: size under the real grid load, and equalize fan-out with dummies
- Calibration is cheap: three MOSCAP banks per stage buy a 6× tighter LSB spread

Future work

- DLL-based background calibration to replace static trim codes
- Voltage-to-time front end: reuse the TDC as an ADC quantizer
- Layout and extracted re-simulation

The design in six numbers

15 ps resolution t_0 , spec asks < 20 ps

0.85 ps LSB corner spread after calibration

11.7 pJ energy per conversion (TT)

0.49 pJ / step figure of merit (TT)

1306 μm total transistor width ΣW

< 1 fs arbiter dead zone

All design, simulation and analysis were carried out jointly by both team members.

Thank you. Questions?

Group 22: Daniel Tyukov, Joris Hoogeweegen
2-D Vernier TDC, EE4615 Digital IC Design II